System Requirements Specification

Smart Hydroponic Grow Kit

Functional Requirements

Category	Requirement ID	Requirement Description	Specification	Verification Method
Monitor	FR-MON-01	Temperature Monitoring	Range: 10-40°C ±0.5°C Update interval: 5 minutes Sensor: Thermocouple/DS18B20	Calibrated thermometer comparison
	FR-MON-02	Humidity Monitoring	Range: 20-90% RH ±3% Update interval: 5 minutes	Salt calibration test
	FR-MON-03	pH Level Monitoring	Range: 4.0-9.0 pH ±0.1 Update interval: 15 minutes	Buffer solution calibration (4.0, 7.0, 10.0)
	FR-MON-04	CO_2 Concentration Monitoring	Range: 400-2000 ppm ±50 ppm Update interval: 30 minutes	Calibration with reference gas
	FR-MON-05	Water Level Monitoring	Reservoir capacity: 30L ±0.5L Low threshold: 20% capacity	Manual measurement verification
	FR-MON-06	Nutrient Concentration (EC)	Range: 0-5.0 mS/cm ±0.1 mS/cm Update interval: 1 hour	Calibration solution verification
Control	FR-CTL-01	LED Lighting Control	PWM control: 0-100% ±5% Schedule: 16/8 photoperiod Intensity: 0-500 µmol/m²/s	Lux meter measurement, schedule verification
	FR-CTL-02	Water Pump Control	Flow rate: 600L/h ±10% Schedule: 15min ON/45min OFF	Flow meter measurement, timer verification

			Manual override capability	
	FR-CTL-03	Fan Speed Control	PWM control: 25-100% speed Auto mode: Temp/Humidity based Manual control available	Anemometer measurement, response testing
	FR-CTL-04	Temperature Regulation	Heating: Nichrome wire 0-100W Cooling: Fan speed modulation Setpoint maintenance: ±1.5°C	Temperature stability test over 24h
	FR-CTL-05	pH Auto-correction	Dosing pumps: 0.1ml increments Adjustment range: 5.5-6.5 pH Delay between adjustments: 15min	pH drift correction test
Alerting	FR-ALT-01	Threshold Breach Alerts	Temperature: <15°C or >30°C pH: <5.5 or >6.5 Water level: <20% CO_2: <400 ppm or >1500 ppm	Simulated breach conditions
	FR-ALT-02	Component Failure Alerts	Pump failure detection Sensor disconnection Communication failure Power interruption	Component disconnection testing
	FR-ALT-03	Alert Delivery Methods	Mobile app notification SMS (South African networks) Email Local audible/visual alarm	Multi-channel delivery test
	FR-ALT-04	Maintenance Reminders	Filter replacement: 30 days pH calibration: 14 days	Schedule-based trigger testing

			Nutrient change: 21 days System cleaning: 60 days	
Data	FR-DAT-01	Local Data Logging	Storage: SD card 8GB minimum Retention: 30 days minimum Format: CSV with timestamp Parameters: All sensor readings	Data retrieval and integrity check
	FR-DAT-02	Cloud Data Upload	Interval: 15 minutes Protocol: MQTT/HTTPS Uptime: >99% Retention: 24h minimum real-time	Network failure/recovery testing
	FR-DAT-03	User Data Access	Mobile app: Real-time dashboard Web interface: Historical graphs Export: CSV download capability API: RESTful endpoints	Cross-platform access testing
	FR-DAT-04	Growth Analytics	DLI calculation VPD computation Growth rate estimation Yield prediction	Algorithm validation with actual growth

Non-Functional Requirements

Category	Requirement ID	Requirement Description	Specification	Verification Method
Reliability	NFR-REL-01	System Uptime	Operational uptime: >99% MTBF: >10,000 hours Mean Time To Repair: <4 hours	Continuous operation stress test
	NFR-REL-02	Sensor Accuracy	Temperature: ±0.5°C Humidity: ±3% RH pH: ±0.1 pH CO_2: ±50 ppm	Calibration against certified references
	NFR-REL-03	Network Reliability	Local connectivity: 99.5% Cloud sync: 99% successful Reconnection: <60 seconds after outage	Network failure simulation
	NFR-REL-04	Power Resilience	Load shedding survival: 4 hours Brownout operation: 180-250V Surge protection: 2kV line-to-line	Power quality simulation testing
Response	NFR-RSP-01	Control Response Time	Sensor to actuation: <2 seconds User command execution: <1 second Alert generation: <30 seconds	Real-time performance measurement

	NFR-RSP-02	User Interface Response	App loading: <3 seconds Data refresh: <5 seconds Graph rendering: <2 seconds	UI/UX performance testing
	NFR-RSP-03	Cloud Communication	Data upload: <5 seconds Command download: <2 seconds Sync interval: 15 minutes	Network latency testing
	NFR-RSP-04	Emergency Response	Critical alert delivery: <60 seconds Safety shutdown: <500ms Backup activation: <1 second	Emergency scenario simulation
Safety	NFR-SAF-01	Electrical Safety	NRCS VC 8035 compliance IP44 enclosure rating Earth leakage protection: 30mA Double insulation requirement	Certified laboratory testing
	NFR-SAF-02	Water-Electricity Separation	SELV circuits near water: <24V DC Waterproof connectors: IP67 rating Drip loops and drainage	Water exposure testing
	NFR-SAF-03	Thermal Safety	Max surface temperature: <60°C Over-temperature	Thermal imaging and testing

			shutdown: 85°C Fire-resistant materials: UL94 V-0	
	NFR-SAF-04	Chemical Safety	Food-grade materials Leachate testing: SAZAS compliant Child-safe reservoir access MSDS documentation provided	Material certification review
Maintainability	NFR-MNT-01	Modular Design	Component replacement: <15 minutes Tool requirements: Basic only Spare part availability: <5 business days	Physical maintenance simulation
	NFR-MNT-02	Software Updates	OTA update capability Rollback functionality Update duration: <10 minutes No data loss during update	Update process testing
	NFR-MNT-03	Calibration Simplicity	pH calibration: <10 minutes Sensor calibration: Guided process Calibration reminders automated	User testing with novices

	NFR-MNT-04	Documentation	Setup guide: <30-minute target Troubleshooting flowchart Multilingual: EN/AF/ZU Video tutorials available	User comprehension testing
Power Efficiency	NFR-PWR-01	Power Consumption	Maximum draw: 150W continuous Average consumption: 80-120W Standby power: <1W Monthly target: <90 kWh	Power meter measurement
	NFR-PWR-02	Energy Management	Load shedding response Peak demand reduction Solar readiness Battery efficiency: >85%	Power management testing
	NFR-PWR-03	Component Efficiency	LED efficiency: >2.5 µmol/J Pump efficiency: >40% PSU efficiency: >85%	Component specification verification
Usability	NFR-USG-01	Setup Time	Initial setup: <2 hours Plant changeover: <30 minutes Configuration: <15 minutes	First-time user testing
	NFR-USG-02	User Interface	Mobile app: 90% task success rate Error recovery: <3 steps Learning curve:	Usability testing with target demographic

			<1 day proficiency	
	NFR-USG-03	Accessibility	Font size adjustable Color contrast compliant Voice control ready Physical buttons for critical functions	Accessibility standard testing
Environmental	NFR-ENV-01	Operating Conditions	Temperature: 15-35°C ambient Humidity: 30-80% non-condensing Altitude: 0-1500m above sea level	Environmental chamber testing
	NFR-ENV-02	Noise Level	Operational noise: <40 dB at 1m Sleep mode: <30 dB Vibration: Minimal detectable	Sound level meter testing
	NFR-ENV-03	Material Sustainability	Recyclable materials: >80% RoHS compliance Local sourcing preference End-of-life plan	Material analysis and sourcing audit
Cost	NFR-CST-01	Unit Cost	Manufacturing cost: <R 4,500 Retail price: R 7,500-R 9,999 Lifetime: 5 years minimum	Bill of Materials analysis
	NFR-CST-02	Operating Cost	Monthly electricity: <R	Operational cost modeling

			350 Monthly consumables: <R 150 Maintenance: <R 500/year	
	NFR-CST-03	ROI Period	Payback period: <24 months Break-even: 200 units Production scale: 1,000 units/year	Financial projection analysis

Specific Requirements

Requirement Type	Local Consideration	Specification	Compliance Standard
Power Quality	Load Shedding Adaptation	4-hour battery backup minimum Brownout operation: 180-250V Surge protection: 2kV	Eskom grid conditions
Connectivity	Network Reliability	Multi-network SIM support Data cost: <R 50/month Offline operation: 7 days	SA network coverage maps
Safety Standards	Legal Compliance	NRCS VC 8035 mandatory SANS 10142 wiring compliance Local certification required	South African National Standards
Water Usage	Conservation Requirements	Recirculation efficiency: >90% Daily consumption: <5L/kWh Drought mode capability	Municipal water restrictions

Local Support	Service Infrastructure	Spare parts: Johannesburg stock Technical support: Local language Warranty: 2 years minimum	Consumer Protection Act

Verification and Validation Summary

Phase	Requirement Coverage	Test Method	Success Criteria
Unit Testing	Individual components	Bench testing	100% component specification met
Integration Testing	System interaction	End-to-end testing	All interfaces functional
Environmental Testing	NFR-ENV series	Chamber testing	Operation within specified ranges
User Acceptance	Usability requirements	Pilot program (10 users)	90% satisfaction rating
Certification Testing	Safety requirements	NRCS-approved lab	Full compliance certification
Field Testing	Reliability requirements	90-day continuous operation	99% uptime achieved